

Pneumonia Navigator

Participant workbook and decision log

Module F · Participant guide

Synthetic teaching data

Instats · August 5, 2026

The facilitator operates Navigator; you co-author the audit.

1. Make a decision before the prepared entry appears.
2. Record the evidence that supports your choice.
3. Name what would make you revise it.
4. Track the downstream consequence of that revision.

Your deliverable

One Navigator entry you can accept, correct, or reject—and defend with evidence.

Manual route (no login)

navigator.tao-rwd.com

Export Markdown at Step 8.

Among adults in a synthetic health-system cohort, what is the effect of pneumococcal vaccination at baseline, versus no vaccination, on 12-month pneumonia hospitalization?

Observed association

Vaccinated risk **7.19%**

Unvaccinated risk **6.43%**

Crude risk difference **+0.76 pp**

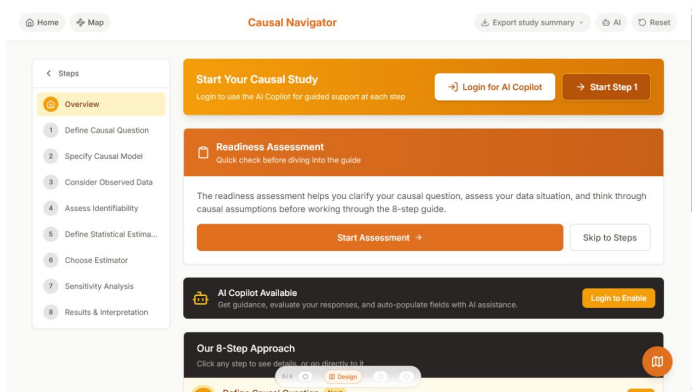
Before interpreting it

Specify the population, treatment, comparator, time zero, outcome window, and contrast.

Synthetic teaching result — not clinical evidence.

The Navigator is an eight-step spine

Overview



1. Question
2. Causal model
3. Observed data
4. Identification
5. Estimand
6. Estimator
7. Sensitivity
8. Interpretation

Each step constrains the next.

Checkpoint 1: specify the causal question

Question

Population

Treatment

Comparator

Time zero

Outcome window

Causal contrast

Revision trigger

Which single phrase in your specification would most change the target estimand?

BASELINE NODES

- age
- prior pneumonia
- prior vaccination
- current vaccination A
- pneumonia hospitalization Y

Model decision

Draw the arrows. Which baseline variables belong in the adjustment set? Why?

OBSERVED DATA

People	5,000
Vaccinated	1,627
Pneumonia events	334
Missingness	none
Censoring	none in fixed data

Mapping decision

Can each Step 1 element map to a variable and time window?

Checkpoint 3: audit identification

Main activity

Assumption	Evidence you would seek	What would make you revise it
Exchangeability		
Positivity		
Consistency		
No interference		

Choose one

Which assumption is weakest in real EHR data? The fixed dataset cannot certify these scientific assumptions.

CAUSAL TARGET

$$E[Y^1] - E[Y^0]$$

STATISTICAL ESTIMAND

$$\begin{aligned}\psi &= E_W\{Q(1, W) - Q(0, W)\}, \\ Q(a, W) &= E(Y \mid A = a, W).\end{aligned}$$

Link

Which Step 4 assumptions identify this target?

PROPOSED ESTIMATOR

TMLE

- $\bar{Q}(A, W)$: outcome regression
- $g(W)$: treatment mechanism
- influence-function interval

Boundary

What can TMLE reduce—and what can it not repair?

Checkpoint 4: plan how the conclusion could fail

Sensitivity

Unmeasured confounding

Could residual frailty or care seeking reverse the direction?

Outcome error

Could pneumonia coding change direction or magnitude?

Selection

Could death, disenrollment, or EHR capture be informative?

Estimation

Do learners or propensity truncation change the result?

Choose one gap

Specify one analysis or source of evidence. A confidence interval does not measure these causal gaps.

PROPOSED STATEMENT A

“In this synthetic exercise, adjustment reverses the crude association under the stipulated causal model.”

PROPOSED STATEMENT B

“The vaccine prevents pneumonia in real patients.”

Accept, correct, or reject

For each claim, cite one assumption or limitation. Rewrite the weaker claim so it says no more than the design supports.

Choose one field from Steps 1–8.

- 1 What evidence makes the entry defensible?
- 2 What would make you revise it?
- 3 Which downstream decision changes if you revise it?

Use the answer key after the exercise

Compare your reasoning with the completed worked example; do not substitute it for your argument.